

VLSI

September 18th, 2026

Due date: September 25th, 2026

Name: _____ DU ID: _____ Section: _____

Lab 1 – Directions

Design a 5-bit R-2R ladder digital to analog converter (DAC).

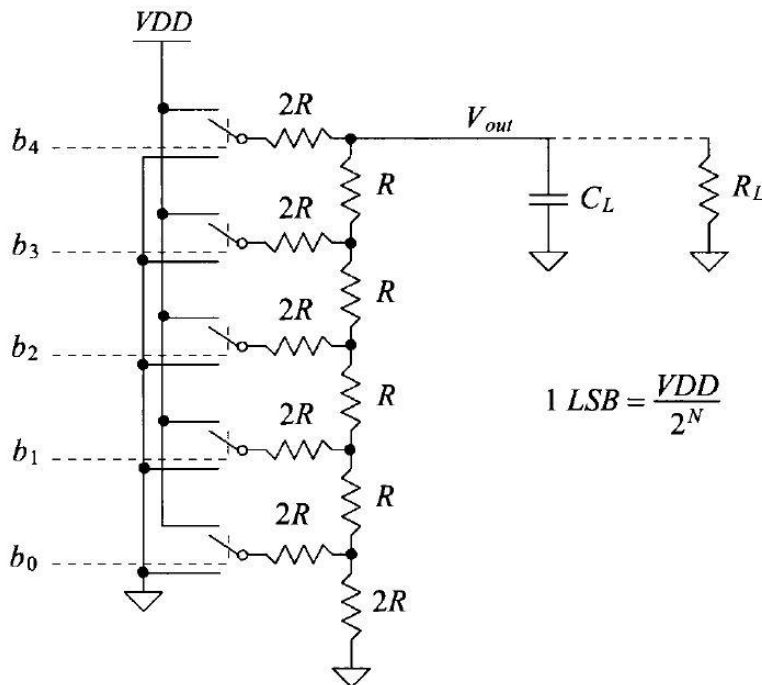


Figure 30.14 Voltage-mode (5-bit) DAC without an op-amp.

Your lab report should document the following:

SCHEMATIC

- The design of a 5-bit DAC using an n-well R of 10k
 - o The $2R$ resistor should be implemented with two separate 10k resistors in series
- Find and explain how to determine the output resistance of the DAC

- Delay, driving a load
 - o Ground all DAC inputs except B4. Connect B4 to a pulse source (0 to VDD) and show, and predict using $0.7RC$, the delay the DAC has driving a 10pF load
 - o Verify the simulation results match your hand calculations
- Simulations to verify your design functions correctly
 - o Apply different values to the input of the DAC and check if the output has the correct voltage
 - o Explain what happens if the DAC drives a 10k load

LAYOUT

- Use the n-well to layout a 10k resistor
 - o Discuss, in your lab report, how to select the width and length of the resistor
- Use this n-well resistor in the layout of your DAC
- Ensure that each resistor in the DAC is laid out in parallel having the same x-position but varying y-positions (the resistors are stacked – serpentine layout)
- All inputs and outputs Pins should be on metal 1
- DRC and LVS your design (show the results in the lab report)
- Zip your final design and report and submit it on canvas

Good luck and have fun.